

# Qatar-2b

R-0682

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-2\_260324 · created 2026-07-27T19:37:17+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2461123.8238 +/- 0.0021 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.086 +/- 0.033                 |
| Transit depth (Rp/Rs)^2                           | 0.75 +/- 0.57 [%]               |
| Orbital Inclination (inc)                         | 89.5 +/- 1.69                   |
| Airmass coefficient 1 (a1)                        | 1.014 +/- 0.016                 |
| Airmass coefficient 2 (a2)                        | -0.011 +/- 0.012                |
| Scatter in the residuals of the lightcurve fit is | 1.56 %                          |
| Best Comparison Star                              | #1 - [105, 444]                 |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.0708 +/- 0.0028               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R* fit vs published    | 0.086 ± 0.033 vs 0.16327 (deviation 2.34σ) |
| Duration fit vs published | 0.0708 d vs 0.0767 d (deviation 2.1σ)      |
| Mid-time O-C              | -5.2 min                                   |
| Depth SNR                 | 2.6                                        |
| Residual scatter          | 1.56 %                                     |

### Light curve

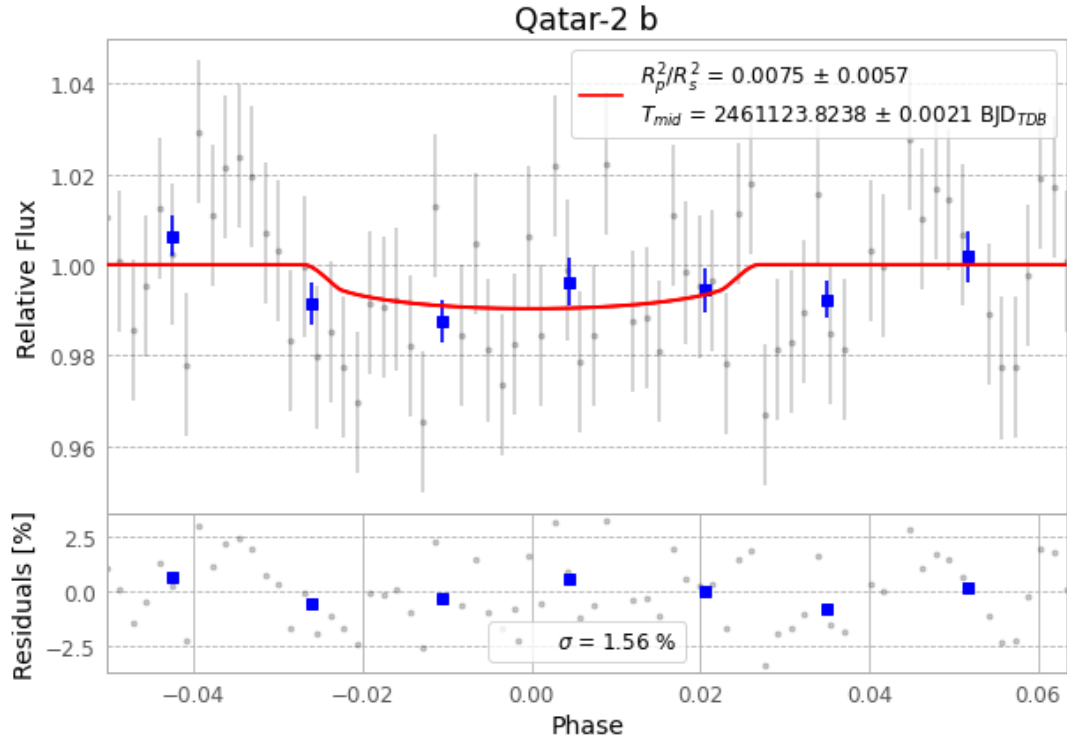


Figure 1 — FinalLightCurve\_Qatar-2 b\_2026-03-23.png (SHA-256 d74f508e952c8cfb...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [210, 477], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 c80653e6adf79ce8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 62f7813

## Data provenance

74 FITS frames (49 MB), Qatar-2260324060016.FITS → Qatar-2260324093916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-260324.log (bcf00ed882a9...), FinalLightCurve\_Qatar-2 b\_2026-03-23.png (d74f508e952c...), FinalLightCurve\_Qatar-2 b\_2026-03-23.csv (392da28da4e3...)

## Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 19:37Z.